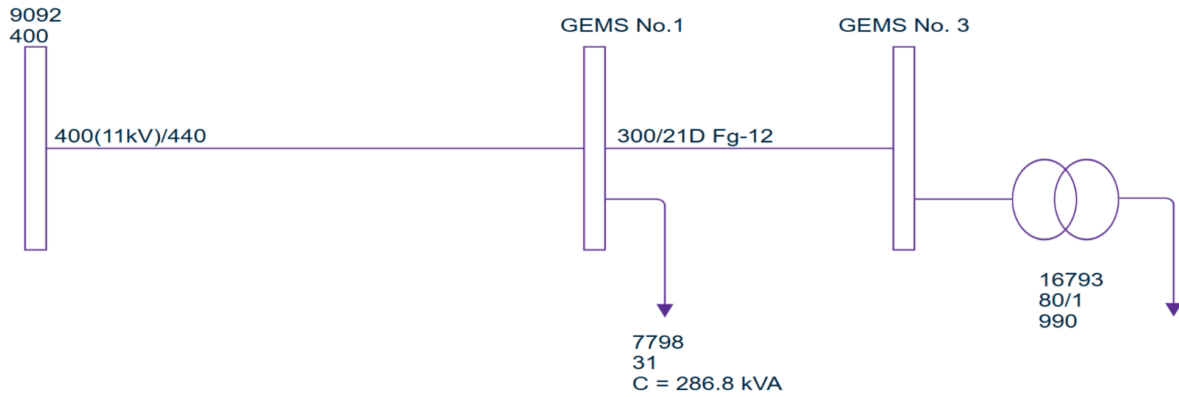


Bus Configuration



CABLE DATA

Source	Destination	Cable length	Cable size
SEEPZ REC-STN (9092)	GEMS No.1 (34205)	20	400
SEEPZ REC-STN (9092)	GEMS No.1 (34205)	148.3	400
SEEPZ REC-STN (9092)	GEMS No.1 (34205)	246.1	400
GEMS No.1 (34204)	GEMS No.3 (16794)	80.9	300
GEMS No.1 (34204)	GEMS No.3 (16794)	92.3	300
GEMS No.1 (07798)	-	10	300
GEMS No.3 (16793)	GEMS No. 3 (DT)	10	300
GEMS No.3 (16792)	SEEPZ EAST (05126)	279.1	300

POLYCAB THREE CORE MV AS/NZS 1429.1 12.7/22 (24) KV

MV Cable with Copper Conductor, XLPE Insulation, Copper Screen and Unarmoured

ELECTRICAL CHARACTERISTICS:

No. of Cores	Core Cross sectional Area	Max. DC Resistance at 20°C	Max. AC Resistance at 90°C	Approx. Capacitance	Approx. Inductance	Approx. Reactance	Continuous Current Rating		
							Buried direct in ground	In a buried duct	In Air
									
No.	mm ²	Ω/km	Ω/km	μF/km	mH/km	Ω/km	Amps		
3	35	0.524	0.668	0.16	0.625	0.196	153	133	170
3	50	0.387	0.494	0.17	0.604	0.190	181	158	204
3	70	0.268	0.342	0.2	0.569	0.179	221	193	253
3	95	0.193	0.246	0.22	0.551	0.173	262	231	304
3	120	0.153	0.196	0.24	0.533	0.167	298	264	351
3	150	0.124	0.159	0.26	0.521	0.164	334	297	398
3	185	0.0991	0.127	0.28	0.509	0.160	377	336	455
3	240	0.0754	0.097	0.31	0.496	0.156	434	390	531
3	300	0.0601	0.078	0.33	0.484	0.152	489	441	606
3	400	0.047	0.062	0.37	0.473	0.149	553	501	696
3	500	0.0366	0.049	0.41	0.462	0.145	632	574	800

*: Current Ratings are based on IEC 60502-2 & IEC 60287, Max. Conductor Temperature at 90°C, Ambient temperature at 30°C in Air / at 20°C in Ground, Thermal resistivity of Soil 1.5 k.m/W & for earthenware ducts 1.2k.m/W and Depth of Laying 0.8m.

Source	Destination	Cable length	Cable size	R (ohm /km)	X (ohm /km)	C (μF/km)
SEEPZ REC-STN (9092)	GEMS No.1 (34205)	246.1	400	0.062	0.149	0.37
GEMS No.1 (34204)	GEMS No.3 (16794)	92.3	300	0.078	0.152	0.33
GEMS No.1 (07798)	-	10	300	0.078	0.152	0.33
GEMS No.3 (16793)	GEMS No. 3 (DT)	10	300	0.078	0.152	0.33

CONSUMER LOAD DATA

Substation	Load PF	CP Load	Connected load	AP Load	Max Demand	voltage	Con. Load
GEMS No.1	97.17	1.79	286.8	25.23	304	440	15.96

TRANSFORMER LOADING DATA

Substation	Rating	%Z	Loading
SEEPZ REC-STN (9092)	990 kVA	17.39	579 kVA

Short-circuit impedance at rated current	
Rated power kVA	Minimum short-circuit impedance %
Up to 630	4,0
631 to 1 250	5,0
1 251 to 2 500	6,0
2 501 to 6 300	7,0
6 301 to 25 000	8,0
25 001 to 40 000	10,0
40 001 to 63 000	11,0
63 001 to 100 000	12,5
Above 100 000	>12,5
NOTE 1 Values for rated power greater than 100 000 kVA are generally subjected to agreement between manufacturer and purchaser.	
NOTE 2 In case of single-phase units connected to form a three-phase bank, the value of rated power applies to three-phase bank rating.	

Results and Discussion

5 Year Load increment results

Case	Load (kW)	Losses (kW)	Reactive Power (kVar)	Reactive Losses(kVar)
Original	901.266	18.28	140.54	66.63
Year 1	902.36	19.21	138.22	68.5
Year 2	921.40	20.36	141.79	72.07
Year 3	941.04	21.58	145.57	75.85
Year 4	961.31	22.87	149.57	79.86
Year 5	982.24	24.25	153.82	84.11

Year 5 results with capacitor banks installed at all the loads (Capacitor at each load is of the size of given capacitor bank)

Case	Capacitor bank size (kVar)	Load (kW)	Losses (kW)	Reactive Power (kVar)	Reactive Losses(kVar)	Voltage at GEMS 1 bus LV (V)	Voltage at GEMS 3 bus LV (V)
Year 5	N/A	982.24	24.25	153.82	84.11	251.43	244.42
Capacitor bank 1	25	982.11	24.05	105.14	83.23	252	245.45
Capacitor bank 2	50	982.02	23.91	56.13	82.66	252.57	246.49
Capacitor bank 3	75	982	23.85	6.78	82.42	253.14	247.53
Capacitor bank 4	100	981.86	23.85	-42.92	82.47	253.72	248.59

- Adding capacitor banks at the load points results in a clear drop in reactive power, especially once the size reaches 50–75 kVar. The active losses also reduce slightly because the current drawn from the network is lower. Voltage levels rise at both the GEMS-1 and GEMS-3 buses, showing better voltage support under compensation.
- When a 100 kVar bank is used, the system becomes over-compensated, which pushes reactive power into the negative. For this reason, the 50–75 kVar range appears to provide the best balance. Overall, the capacitors improve both power factor and voltage without overcorrecting.

Year 5 results with Diesel Generators installed at all the loads (Each load has a rating of given generator/2)

Case	Generator Capacity (kW)	Load (kW)	Losses (kW)	Reactive Power (kVar)	Reactive Losses(kVar)
Year 5	N/A	982.24	24.25	153.82	84.11
Generator 1	250	723.36	15.35	119.66	49.94
Generator 2	500	466.59	8.61	94.96	25.24
Generator 3	750	211.96	3.95	79.33	9.61
Generator 4	1000	-40.71	-1.26	72.43	2.71

- Introducing diesel generators drastically reduces the load seen by the grid, especially once the DG size reaches 500–750 kW. This leads to a sharp fall in losses, dropping from around 24 kW to just 4–9 kW depending on the DG size.
- However, using a very large DG (1000 kW) causes reverse power flow, which may not be ideal for the network.

Year 5 results with Solar PV installed at all the loads (Each load has a rating of given PV array/2)

Case	Solar PV size (kW)	Inverter size (1.1 * Solar PV size) (kVA)	Load (kW)	Losses (kW)	Reactive Power (kVar)	Reactive Losses(kV ar)
Year 5	N/A	N/A	982.24	24.25	153.82	84.11
Solar PV 1	250	275	769.36	15.67	134.09	53.26
Solar PV 2	500	550	512.47	8.8	108.24	27.42
Solar PV 3	750	825	257.74	4.05	91.51	10.69
Solar PV 4	1000	1100	4.96	1.26	83.54	2.72

- Solar PV produces effects very similar to DG but without causing reverse flow, even at the 1000 kW size. As PV generation increases, the load on the network drops sharply, which brings the losses down from 24.25 kW to as low as 1.26 kW.
- Reactive losses fall as well, reflecting the lower current in the lines. The reduction in feeder current also minimizes voltage drops, improving the voltage profile across the network. Since the generation is distributed, the impact on losses is even more pronounced than with DG. Overall, PV provides the strongest improvement in losses and network loading among all the options.